SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – SHORT ANSWER TEST

Course Code:	CSA16	**Course Title:**	Data Warehousing and Data Mining
CO Assessed:	CO1 – Precision (BL1, BL2)	**Assessment:**	Assessment Tool 1 – Short Answer Test
Weightage:	10% of CIA	**Total Marks:**	100 (5 Questions × 20 Marks)
CO1 Statement:	*Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2*		
Student Name:	S. Mythili	**Reg. No.:**	192411250
Section / Batch:	-	**Date:**	15/7/26

Instructions to Students

- Answer all questions. Each question carries 20 Marks. Total: 100 Marks.
- Clearly show all requirements, process, operations.
- Use appropriate models, schema represented scenario wise
- Justify each step in different dimension specified and measures clearly.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Short Answer Test	Architecture of Data Warehouse, ETL, OLAP	Q1 – Q5	Q1 –20 Q2 –20 Q3 –20 Q4 –20 Q5 –20
		GRAND TOTAL:	100 Marks

SECTION A – SHORT ANSWER TEST

1. Suppose that a data warehouse for City Hospital consists of the following four dimensions: patient, doctor, department, and time, and two measures visit_count and avg_charge. At the lowest conceptual level, avg_charge stores the actual consultation fee charged for a patient visit. At higher conceptual levels, it stores the average fee.

(a) Draw a snowflake schema diagram for the data warehouse.

(b) Starting with the base cuboid [patient, doctor, department, time], what specific OLAP operations should be performed to list the average consultation fee department-wise for each year?

(c) If each dimension has four levels (including all), how many cuboids will this cube contain (including base and apex cuboids)?

2. A smart city project is collecting data from multiple sources, including:

- Traffic sensors for vehicle counts and congestion levels.
- IoT devices for monitoring air quality, noise levels, and energy consumption.
- Public transportation systems for ridership data and punctuality.
- Citizen feedback from a mobile app.

Questions:

Fact and Dimension Design:

- Design a star schema to analyze traffic congestion patterns, energy consumption trends, and public transportation efficiency.
- What measures and dimensions would you include? Justify your design.

3. Suppose that a banking data warehouse contains two fact tables: Transaction_Fact and Loan_Fact, sharing dimensions customer, branch, time, and account_type.

- Transaction_Fact → transaction_count, amount
- Loan_Fact → loan_amount, interest

(a) Draw a galaxy schema diagram.

(b) Identify OLAP operations to obtain branch-wise average transaction amount and loan amount yearly.

(c) If each dimension has four levels, calculate total cuboids.

4. A manufacturing company monitors product quality and production efficiency. The data warehouse contains:

- Dimensions:
  - Product: Category, type, and batch.
  - Factory: Location, department, and line number.
  - Time: Year, quarter, month, and week.
  - Supplier: Region, rating, and contract type.
- Measures:
  - Defect rate.
  - Production units.
  - Supplier reliability percentage.

Questions:

- Roll-up: Summarize defect rates by product category for the current year across all factories.
- Drill-down: Focus on a specific factory to analyze defect rates by department and product type.
- Slice: Isolate production unit data for suppliers with a rating of 4+ across all regions.
- Dice: Analyze defect rates and supplier reliability for selected product categories and factory locations.

5. A retail chain manages inventory and sales data across multiple stores. The data warehouse contains:

- Dimensions:
  - Product: Category, subcategory, and SKU.
  - Store: Region, city, and store ID.
  - Time: Year, quarter, month, and day.
  - Customer: Membership tier, age group, and gender.
- Measures:
  - Inventory turnover.
  - Total sales revenue.
  - Units sold.

Questions:

1. Roll-up: Summarize inventory turnover by product category across all stores for the last quarter.
2. Drill-down: Focus on sales data for electronics in a specific region, analysing by subcategory and SKU.
3. Slice: Isolate sales revenue data for male customers in the Bronze membership tier across all stores.
4. Dice: Analyze sales revenue and units sold for selected cities and product categories during specific time periods.

Code of Conduct

I S.Mythili (192411250) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Mythili

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Marks Evaluation:

Question No.	Understanding of Concepts (25%)	Explanation (25%)	Application (20%)	Organization (15%)	Accuracy (10%)	Clarity (5%)	Total Marks
Q1 (20 Marks)							
Q2 (20 Marks)							
Q3 (20 Marks)							
Q4 (20 Marks)							
Q5 (20 Marks							
					Overall Total (100)		

Faculty In-charge	Course Coordinator	Head of Department
Signature: ______________	Signature: ______________	Signature: ______________
Date: ________________	Date: ________________	Date: ________________

CSA1613 → Data Warehousing and Data Mining : Q. Mythili
192411250

Assessment - 1

1. City Hospital Data Warehouse :

(a) Snowflake Schema Diagram

Patient_Dim

Patient_ID (PK)
Name
Age
Gender
City

Doctor_Dim --- Visit_fact ---
Department_Dim

Doctor_ID (PK)
Doctor_Name
Specialization
experience

Patient_ID (FK)
Doctor_ID (FK)
Dept_ID (FK)
Time_ID (FK)

Dept_ID (PK)
Dept_Name
Hospital

Time_Dim
Time_ID (PK)
Day
Month
Quater
Year

(b) OLAP operations:

Goal → find The average consultation fee department - wise for each year

Step 1 → Roll up time dimension from Day → Month → Quarter → Year

2 → Roll up department dimension

3 → Aggregate

SQL - like Result

Year	Dept.	Avg. Charge
2024	Cardiology	₹650
2024	orthopedics	₹ 550
2025	Cardiology	₹700

(c) Number of Cuboids :

for n dimensions having L levels,

Cuboids = L^n

dimensions = 4

levels per dimension = 4

$4^4 = 256$.

Smart City Project:

```
                Time
                  |
location -- fact -- sensor
        |
    transport
        |
     Citizen
```

Measures → Vehicle count, congestion index
energy consumption, Passenger count, delay time
air Quality index, Noise level.

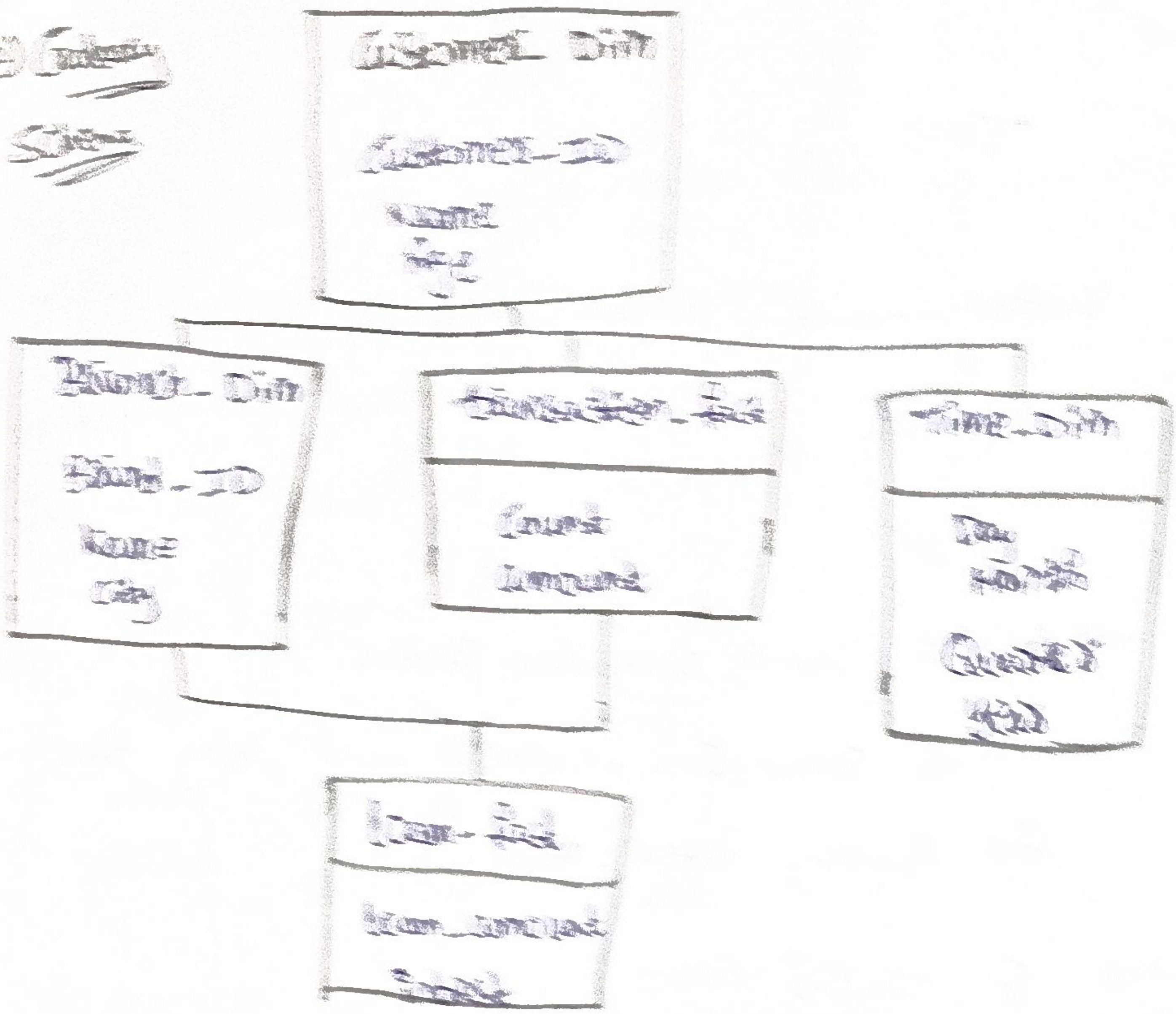
Dimensions & Justification:

Time → Daily / Monthly trend analysis
location → compare city zones
transport → analyze buses, metro, trains
Sensor → Monitor Pollution & energy
Citizen → analyze feedback by user groups

This schema enables traffic, Pollution, energy & transport analytics.

=> Banking Data Warehouse :

(a) Galaxy Schema



b) OLAP operations

Branch	Year	Avg Transaction	Avg Loan
Hyd	2025	₹20,500	₹5,00,000
Warangal	2025	₹15,200	₹4,80,000

(c) Total Cuboids :

dimensions = 4

levels = 4

$4^4 = 256$

256 Cuboids.

1. Manufacturing Company:

Roll-UP

Requirement → Summarize defect rates by Product category for current year across all factories

Operation

* Roll-UP factory → all factories
* Roll-UP Product → category
* Roll-UP Time → current year

Drill-down

Focus on one factory

factory

Factory A

Then

department

↓

Production line

↓

Product type

Ex:

Factory A

→ assembly

→ line 2

→ laptop

defect rate = 1.5%

Slice:

Condition

Supplier Rating ≥ 4

across all regions

Result

Supplier	Rating	Production units
ABC	5	15,000
XYZ	4	12,500

5. **Retail Chain Data Warehouse:**

Roll-UP

Requirement → inventory turnover by category across all stores during last quarter

Operations

Store → City → Region → all stores

Product SKU → Subcategory → category

Time Day → Month → Quarter

Drill-down

Requirement

electronic sales

Region

South

Then

Category

↓

Subcategory

↓

SKU

Ex:

electronics

→ Mobile phones

→ Samsung galaxy

Sales = ₹4,50,000

Slice:

condition

Customer Membership = Bronze

Measure

Sales Revenue

Store	Bronze Revenue
Hyd	₹ 5,00,000
Chennai	₹ 4,50,000

Dice:

→ Cities = Hyd, Chennai

→ Categories = electronics, Grocery

→ Time = January - March